

Section 1 — Physics and Human Society

1. The branch of science that deals with matter and energy and their interactions is

(a) physics (b) botany (c) sociology (d) geology

3. Computer tomography (CT) and magnetic resonance imaging (MRI) are examples of

(a) athletic training programmes (b) medical techniques made possible by physics (c) financial audit procedures (d) methods of food preservation

5. Automobiles, airplanes and space shuttles could not be constructed and operated without

(a) the help of physics experts (b) astronomers alone (c) historians alone (d) economists alone

7. Satellite telephones and television broadcasting work by

(a) sending messenger balloons (b) transmitting and receiving electromagnetic waves (c) bouncing sound waves off clouds (d) carrying signals through underground pipes

9. The internet, smartphones and computers are possible because of technologies built on

(a) principles of classical poetry (b) the study of ancient inscriptions (c) rules of international trade (d) principles of physics such as electricity and electromagnetism

11. Designing energy systems such as electric power grids and hydroelectric dams relies chiefly on knowledge of

(a) horticulture (b) calligraphy (c) physics (d) numismatics

13. The strongest everyday example that physics is useful in daily life is

(a) the lettering on a coin (b) the flavour of a meal (c) the weave of a carpet (d) the electricity that lights and heats a home

2. Physics improves the quality of human life mainly by

(a) drafting government rules (b) organising cultural festivals (c) digging irrigation canals (d) providing the understanding needed to build new instruments and technologies

4. The design and safe flight of an airplane depend chiefly on

(a) the language used by the pilot (b) the paint colour of the aircraft (c) physical principles such as those governing lift, thrust and drag (d) the number of seats in the cabin

6. Physicists have contributed to defence and security mainly by

(a) designing hospital wards (b) mapping tourist trails (c) organising training drills (d) informing the design of weapons for modern defence

8. Which of the following is NOT a physics-based contribution to health care?

(a) laser surgery (b) ultrasonic imaging (c) selecting disease-resistant crop seeds (d) positron emission tomography (PET)

10. A solar panel converts sunlight into electrical energy. This technology depends on the physics of

(a) light and the way electrons behave in materials (b) ocean current motion (c) sound reflection off surfaces (d) soil erosion processes

12. Telescopes, microscopes, spectrometers and CT scanners show that physics has provided

(a) equipment used only in sports (b) scientific instruments applied across many disciplines (c) purely decorative tools (d) nothing beyond the physics laboratory

14. Physics contributes to a country's economy mainly through

(a) technologies such as electricity generation and industrial machinery (b) printing currency notes (c) charging customs duties (d) composing national anthems

15. Which pair of home devices depends directly on the physics of heat and electricity?

- (a) broom and dustpan (b) pillow and blanket (c) refrigerator and electric iron (d) table and chair

Section 2 — Development of Physics and Contributions of Physicists

16. The word “physics” comes from the Greek word “physis”, which means

- (a) nature (b) strength (c) motion (d) study

18. Galileo Galilei is commonly regarded as the father of

- (a) geometry (b) taxonomy (c) cartography (d) modern experimental science

20. The special and general theories of relativity were developed by

- (a) Albert Einstein (b) Robert Hooke (c) Henry Cavendish (d) Johannes Kepler

22. Early pioneering research on radioactivity was carried out by

- (a) Charles Darwin (b) Edwin Hubble (c) Gregor Mendel (d) Marie Curie

24. Galileo helped overturn reliance on authority in science by insisting that scientific claims must be

- (a) tested by experiment and observation (b) kept secret from other scholars (c) decided by popular vote (d) copied from ancient books

26. The ancient astronomers of Egypt, Babylon and Greece contributed to the early growth of physics by

- (a) carefully observing and recording the motions of the Sun, Moon and stars (b) inventing the electric motor (c) discovering X-rays (d) writing the first tax codes

28. The electron was discovered through experiments on cathode rays carried out by

- (a) Newton (b) Kepler (c) J.J. Thomson (d) Aristotle

17. In ancient Greece, the systematic study of nature, which later grew into physics, is associated with philosophers such as

- (a) Albert Einstein (b) Isaac Newton (c) Aristotle (d) Max Planck

19. The three laws of motion and the law of universal gravitation were formulated by

- (a) James Clerk Maxwell (b) Isaac Newton (c) Michael Faraday (d) Werner Heisenberg

21. James Clerk Maxwell’s greatest contribution to physics was

- (a) discovering the circulation of blood (b) proposing the periodic table (c) unifying electricity and magnetism into a theory of electromagnetism (d) mapping the far side of the Moon

23. Quantum mechanics is the branch of physics that describes

- (a) the motion of the planets (b) the behaviour of atoms and subatomic particles (c) the flow of rivers (d) the formation of fossils

25. Newton’s law of universal gravitation explains

- (a) why the sky appears blue (b) why a bell cannot ring in a vacuum (c) why objects fall to Earth and why planets orbit the Sun (d) why sugar dissolves in water

27. Maxwell’s theory of electromagnetism predicted that light is

- (a) a form of sound (b) an electromagnetic wave (c) a stream of tiny droplets (d) a chemical element

29. Max Planck’s idea that energy is emitted and absorbed in discrete packets, called quanta, became the foundation of

- (a) classical mechanics (b) oceanography (c) cartography (d) quantum physics

30. Einstein explained the photoelectric effect by proposing that light energy travels as

- (a) individual packets of energy called photons (b) a continuous wave that can carry any amount of energy (c) trails of heated gas (d) currents of pure electric charge

Section 3 — Making Physics Knowledge, Communities, Careers and Current Status

31. The ordered series of steps used to gain knowledge based on experimental evidence is called the

- (a) intuitive method (b) scientific method (c) method of authority (d) legal method

33. Obtaining knowledge by seeing, touching, hearing and smelling is called

- (a) sensory perception (b) introspection (c) recollection (d) testimony

35. Concerning testimony as a source of knowledge, the textbook states that written sources are usually

- (a) less reliable than oral sources (b) more reliable than oral sources (c) exactly as reliable as oral sources (d) never reliable at all

37. A stated objective of the Ethiopian Physical Society (EPS) is to

- (a) collect import duties (b) manage the civil service (c) promote physics education and research in the country (d) organise the national lottery

39. Which of the following is a typical physics-related career?

- (a) bookkeeper (b) tour guide (c) librarian (d) medical physicist

41. A planet that orbits a star other than our Sun is called

- (a) a comet (b) an asteroid (c) an exoplanet (d) a nebula

43. The ripples in the fabric of spacetime first detected in 2015 by the LIGO observatories are called

32. In the scientific method, a tentative explanation that can be tested by experiment is a

- (a) conclusion (b) law (c) calculation (d) hypothesis

34. When a patient describes a pain that only the patient feels, the doctor must rely on the patient's

- (a) sensory perception of the doctor's pain (b) laboratory readings (c) introspection (d) recalled dreams

36. Physics communities are organised groups of people who come together

- (a) for scientific purposes (b) to compete for commercial profit (c) to organise political campaigns (d) to stage cultural festivals

38. According to the textbook, physics communities can help their members by providing things such as

- (a) tax exemptions (b) training, scholarships, equipment and computing facilities (c) lifetime free housing (d) driving licences

40. Meteorologists, geophysicists and astronomers base their professional work mainly on the science of

- (a) physics (b) literature (c) law (d) theology

42. A black hole is an object whose gravitational pull is so strong that

- (a) it emits only sound waves (b) nothing, not even light, can escape it (c) it shines more brightly than the Sun (d) it floats above the plane of the solar system

44. The particle whose discovery at CERN in 2012 helped explain how other particles acquire mass is the

- (a) Higgs boson (b) electron (c) neutron (d) proton

- (a) seismic waves (b) sound waves (c) ocean waves
(d) gravitational waves

45. The James Webb Space Telescope was designed to observe the universe mainly in the

- (a) audible range (b) visible range only (c) infrared
(d) alpha-ray range

Section 4 — Measurement, Scientific Notation and Units

46. Measurement is important in physics mainly because it turns observations into

- (a) decorative figures (b) opinions to be debated (c) secret formulas (d) exact quantitative data that can be compared and tested

48. The SI base unit of length is the

- (a) kilometre (b) centimetre (c) metre (d) inch

50. In SI, temperature is measured in

- (a) degrees Celsius (b) degrees Fahrenheit (c) joules
(d) kelvin

52. The SI unit of force is the

- (a) joule (b) pascal (c) newton (d) watt

54. The number 602 000 000 000 000 000 000 000 is best written in scientific notation as

- (a) 6.02×10^{22} (b) 602×10^{20} (c) 60.2×10^{23} (d) 6.02×10^{23}

56. A measurement that lies very close to the true value is described as

- (a) accurate (b) precise (c) rounded (d) random

58. To measure the internal diameter of a narrow tube to a fraction of a millimetre, the most suitable instrument is

- (a) a metre rule (b) a thermometer (c) a stopwatch
(d) a vernier calliper

60. When reading the volume of a liquid in a measuring cylinder, reading the scale at the bottom of the meniscus at eye level avoids

- (a) zero error (b) rounding error (c) calibration error (d) parallax error

47. The science of measurement is called

- (a) metrology (b) meteorology (c) mineralogy (d) taxonomy

49. The SI base unit of mass is the

- (a) gram (b) kilogram (c) newton (d) tonne

51. Electric current is measured in

- (a) amperes (b) volts (c) ohms (d) watts

53. The number 0.0045 is correctly written in scientific notation as

- (a) 4.5×10^2 (b) 4.5×10^{-3} (c) 4.5×10^3 (d) 45×10^{-2}

55. The number of significant figures in 0.00450 is

- (a) 2 (b) 3 (c) 4 (d) 5

57. Several readings that agree closely with one another, even though they may all be far from the true value, are said to be

- (a) accurate (b) exact (c) precise (d) calibrated

59. In SI, the prefix “mega” represents the factor 10^6 , so one megawatt (MW) equals

- (a) 10^6 W (b) 10^3 W (c) 10^{-6} W (d) 10^9 W